STACKS, DEFORMATION THEORY: LEMMA-CANONICAL-CLASS-OBSTRUCTION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5574-5589 of sources/stacks-defos/pinned-original-source.tex; proof span is lines 5591-5712, the complete author proof (120 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 2 of queue_defos.json (pb=4516, stmt 5574-5589, proof 5591-5712).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-canonical-class-obstruction' also exists in stacks-snapshot-20261011 — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 05QA: sources/stacks-defos/pinned-original-source.tex, source lines 5712–5712. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DIU: sources/stacks-defos/pinned-original-source.tex, source lines 5408–5429. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DIV: sources/stacks-defos/pinned-original-source.tex, source lines 5530–5543. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{O} \to \mathcal{O}_0$ be a surjection of sheaves of rings whose kernel is an ideal sheaf $\mathcal{I}$ of square zero. For every object K_0 in $D^-(\mathcal{O}_0)$ the following are equivalent*

1. *the class $\omega(K_0) \in \mathrm{Ext}^2_{\mathcal{O}_0}(K_0, K_0 \otimes_{\mathcal{O}_0} \mathcal{I})$ constructed in Lemma 0DIU is zero,*
2. *there exists $K \in D^-(\mathcal{O})$ with $K \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{O}_0 = K_0$ in $D(\mathcal{O}_0)$.*

Date: stacks-project:defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let K be as in (2). Then we can represent K by a bounded above complex $\mathcal{F}^\bullet$ of flat $\mathcal{O}$-modules. Then $\mathcal{F}_0^\bullet = \mathcal{F}^\bullet \otimes_{\mathcal{O}} \mathcal{O}_0$ represents K_0 in $D(\mathcal{O}_0)$. Since $d_{\mathcal{F}^\bullet} \circ d_{\mathcal{F}^\bullet} = 0$ as $\mathcal{F}^\bullet$ is a complex, we see from the very construction of $\omega(K_0)$ that it is zero.

Assume (1). Let $\mathcal{F}^n$, $d : \mathcal{F}^n \to \mathcal{F}^{n+1}$ be as in the construction of $\omega(K_0)$. The nullity of $\omega(K_0)$ implies that the map

$$\omega = d \circ d : \mathcal{F}_0^\bullet \longrightarrow (\mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I})[2]$$

is zero in $D(\mathcal{O}_0)$. By definition of the derived category as the localization of the homotopy category of complexes of $\mathcal{O}_0$-modules, there exists a quasi-isomorphism $\alpha : \mathcal{G}_0^\bullet \to \mathcal{F}_0^\bullet$ such that there exist $\mathcal{O}_0$-modules maps $h^n : \mathcal{G}_0^n \to \mathcal{F}_0^{n+1} \otimes_{\mathcal{O}} \mathcal{I}$ with

$$\omega \circ \alpha = d_{\mathcal{F}_0^\bullet \otimes \mathcal{I}} \circ h + h \circ d_{\mathcal{G}_0^\bullet}$$

We set

$$\mathcal{H}^n = \mathcal{F}^n \times_{\mathcal{F}_0^n} \mathcal{G}_0^n$$

and we define

$$d' : \mathcal{H}^n \longrightarrow \mathcal{H}^{n+1}, \quad (f^n, g_0^n) \longmapsto (d(f^n) - h^n(g_0^n), d(g_0^n))$$

with obvious notation using that $\mathcal{F}_0^{n+1} \otimes_{\mathcal{O}_0} \mathcal{I} = \mathcal{F}^{n+1} \otimes_{\mathcal{O}} \mathcal{I} = \mathcal{I}\mathcal{F}^{n+1} \subset \mathcal{F}^{n+1}$. Then one checks $d' \circ d' = 0$ by our choice of h^n and definition of ω. Hence $\mathcal{H}^\bullet$ defines an object in $D(\mathcal{O})$. On the other hand, there is a short exact sequence of complexes of $\mathcal{O}$-modules

$$0 \to \mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I} \to \mathcal{H}^\bullet \to \mathcal{G}_0^\bullet \to 0$$

We still have to show that $\mathcal{H}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{O}_0$ is isomorphic to K_0. Choose a quasi-isomorphism $\mathcal{E}^\bullet \to \mathcal{H}^\bullet$ where $\mathcal{E}^\bullet$ is a bounded above complex of flat $\mathcal{O}$-modules. We obtain a commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathcal{E}^\bullet \otimes_{\mathcal{O}} \mathcal{I} & \longrightarrow & \mathcal{E}^\bullet & \longrightarrow & \mathcal{E}_0^\bullet & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle \delta} & & \\
0 & \longrightarrow & \mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I} & \longrightarrow & \mathcal{H}^\bullet & \longrightarrow & \mathcal{G}_0^\bullet & \longrightarrow & 0
\end{array}$$

We claim that δ is a quasi-isomorphism. Since $H^i(\delta)$ is an isomorphism for $i \gg 0$, we can use descending induction on n such that $H^i(\delta)$ is an isomorphism for $i \geq n$. Observe that $\mathcal{E}^\bullet \otimes_{\mathcal{O}} \mathcal{I}$ represents $\mathcal{E}_0^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I}$, that $\mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I}$ represents $\mathcal{G}_0^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I}$, and that $\beta = \delta \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathrm{id}_{\mathcal{I}}$ as maps in $D(\mathcal{O}_0)$. This is true because $\beta = (\alpha \otimes \mathrm{id}_{\mathcal{I}}) \circ (\delta \otimes \mathrm{id}_{\mathcal{I}})$. Suppose that $H^i(\delta)$ is an isomorphism in degrees $\geq n$. Then the same is true for β by what we just said and Lemma 0DIV. Then we can look at the diagram

$$\begin{array}{ccccccccccc}
H^{n-1}(\mathcal{E}^\bullet \otimes_{\mathcal{O}} \mathcal{I}) & \longrightarrow & H^{n-1}(\mathcal{E}^\bullet) & \longrightarrow & H^{n-1}(\mathcal{E}_0^\bullet) & \longrightarrow & H^n(\mathcal{E}^\bullet \otimes_{\mathcal{O}} \mathcal{I}) & \longrightarrow & H^n(\mathcal{E}^\bullet) \\
\downarrow{\scriptstyle H^{n-1}(\beta)} & & \downarrow & & \downarrow{\scriptstyle H^{n-1}(\delta)} & & \downarrow{\scriptstyle H^n(\beta)} & & \downarrow \\
H^{n-1}(\mathcal{F}_0^\bullet \otimes_{\mathcal{O}} \mathcal{I}) & \longrightarrow & H^{n-1}(\mathcal{H}^\bullet) & \longrightarrow & H^{n-1}(\mathcal{G}_0^\bullet) & \longrightarrow & H^n(\mathcal{F}_0^\bullet \otimes_{\mathcal{O}} \mathcal{I}) & \longrightarrow & H^n(\mathcal{H}^\bullet)
\end{array}$$

Using Homology, Lemma 05QA we see that $H^{n-1}(\delta)$ is surjective. This in turn implies that $H^{n-1}(\beta)$ is surjective by Lemma 0DIV. Using Homology, Lemma 05QA again we see that $H^{n-1}(\delta)$ is an isomorphism. The claim holds by induction, so δ is a quasi-isomorphism which is what we wanted to show. □